

Bare Metal & On-Premises Infrastructure with VMware ESXi (Jazz/Ericsson)

Client: Jazz Pakistan

Employer: Ericsson (Performance Team)

Location: Jazz Data Center (On-site)

Project Background

The project aimed to support Jazz's mobile network performance analysis by developing tools to process and evaluate daily KPI data. Our performance team built systems to scan and compare over 12,000 XML files, each representing a network tower across Pakistan. Alongside my primary role in infrastructure management, I also contributed to developing these tools.

Key deliveries included the Delta Scanner (for detecting daily sensor discrepancies), the Baseline Comparison module (for evaluating performance against Jazz's KPIs), and the Huw Cell Comparator (for comparing two network cells).

These were among several tools delivered, and we provided actionable insights through Excel reports with graphs and interactive HTML/CSS dashboards.

Infrastructure Responsibilities

At Jazz (a major network operator and Ericsson's client), I worked onsite in their data center, managing a fully on-premise infrastructure due to strict data privacy policies. I was responsible for provisioning and managing physical servers using **VMware ESXi**, **vCenter**, and **HPE iLO** for remote access. This included configuring **RAID disks**, performing OS installations, managing the full server lifecycle, and backing up/restoring VMware VMs using **OVF files** to ensure business continuity.

I also managed physical networking by connecting bare metal HPE servers via physical network switches and configured **VMware ESXi virtual switches** to handle VM networking and traffic segmentation within the virtual environment.

For monitoring and automation, I performed cluster health checks using **SNMP-based tools** and **Ansible Galaxy VMware modules**, automated VM provisioning, OS configuration, and software deployments with **Ansible**, and developed custom Python scripts to monitor Windows server resources such as disk, CPU, and RAM ([GitHub Demo](#)). Additionally, I built a VMware automation demo project using Ansible ([GitHub Demo](#)).

I collaborated with **Evernex** to physically replace failed hardware components such as hard drives, RAM, and power supplies in HPE servers, minimizing downtime and maintaining operational stability.

All deployment and management tasks were performed onsite at the Jazz data center. (You can get a glimpse of the setup [here](#)).